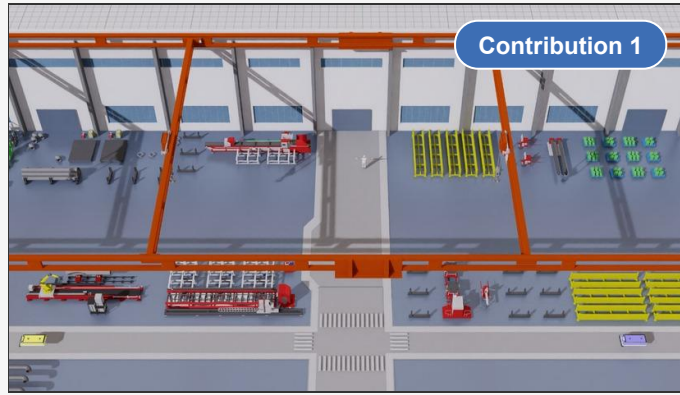
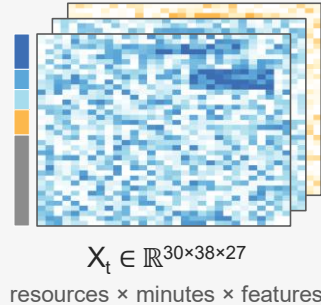
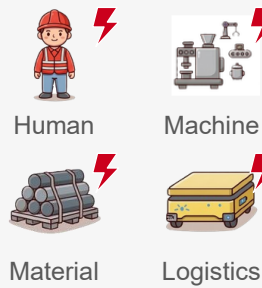


Digital twin & multi-entity data

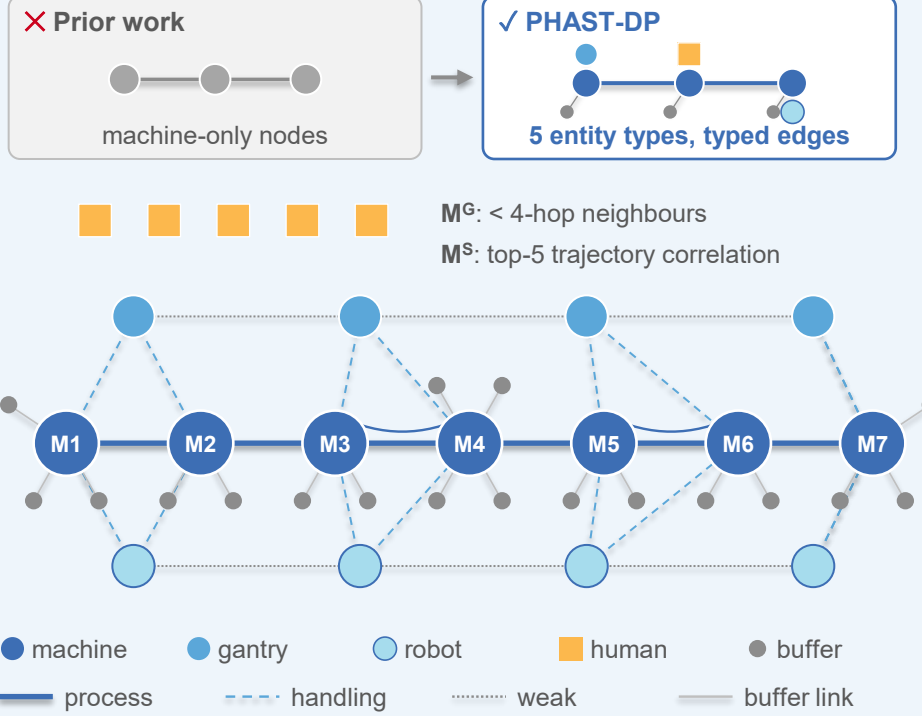


Isaac Sim line · single-axis disturbances

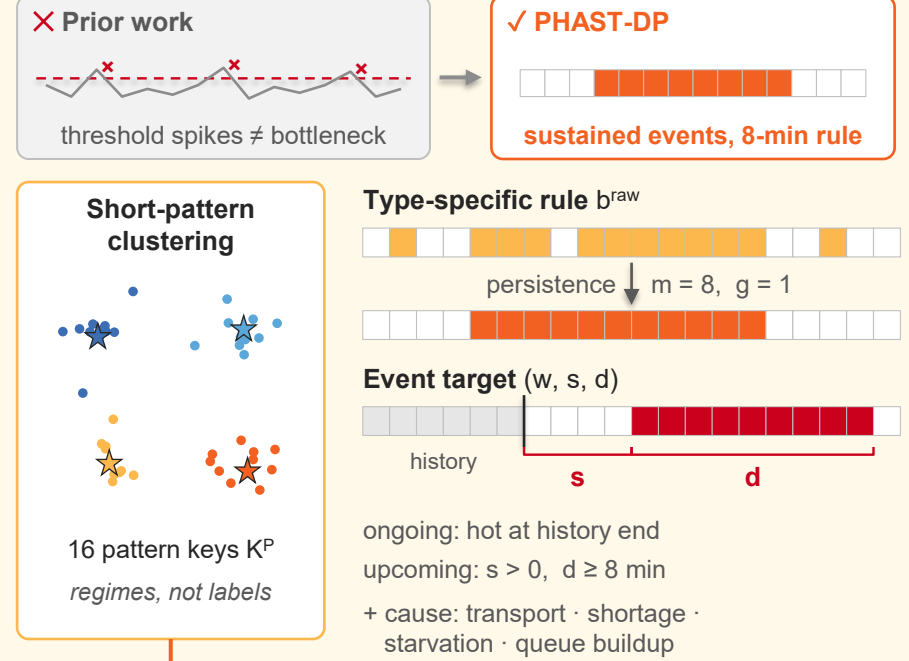


Contribution 1

Contribution 1 · Multi-entity heterogeneous graph



Contribution 2 · Process-informed event labels



PHAST-DP

X_t

G, M^G, M^S

pattern keys K^P

training targets

Grouped embedding

Contribution 1

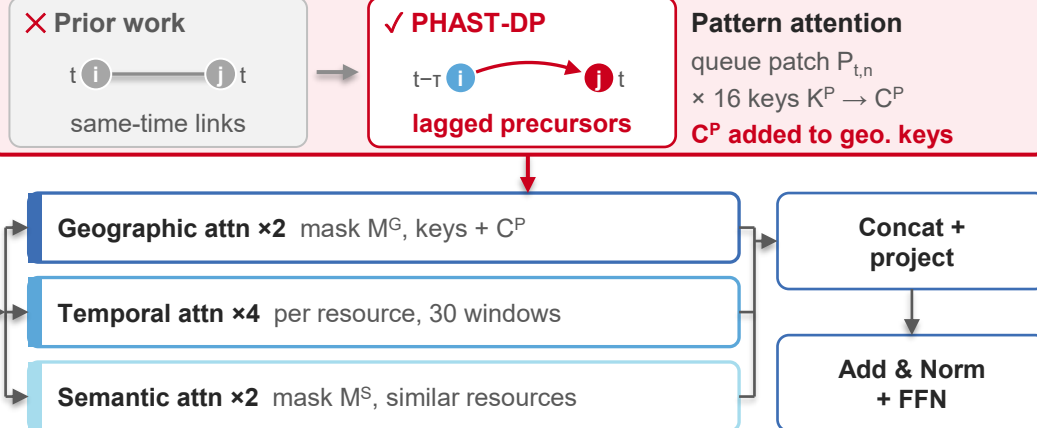
Queue	4
Stall	6
Logistics	4
Shortage	1
Context	7
Type	5

mean

$$+ E_{time} + E_{lap} + E_{node} \rightarrow H^{(0)}$$

Encoder block $\times 5$ ($d = 80$)

Contribution 3 · Delay-propagation module



Multi-task heads

Occurrence p^{cont} / p^{onset}

station event

Start π^s · duration d^d

when and how long

Prefix p^s · occupancy $\hat{O}$

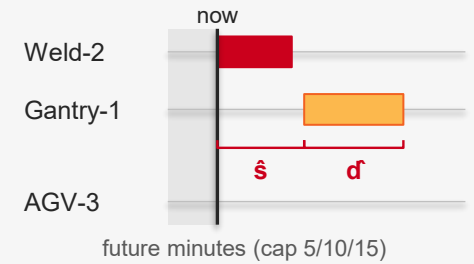
dense future grid

Cause $\hat{c}$ · remaining ℓ

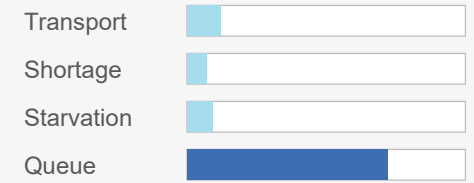
graph-level pooling

validation-controlled decoding

Bottleneck report



Process cause $\hat{c}$



Remaining order time ℓ

